



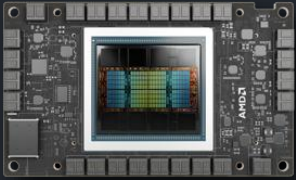
AMD AI & HPC Technology

Sonya Yang

AMD 
together we advance_

End-to-end AI infrastructure

Leadership compute engines for AI

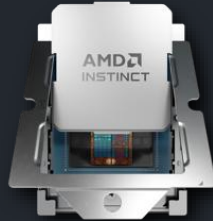


Cloud

Leadership accelerator
for generative AI, LLMs
and inferencing

AMD Instinct™ MI300X
Accelerators

AMD Instinct™ MI325X
Accelerators



HPC

World's first data center
APU for HPC and AI

AMD Instinct™ MI300A
Accelerators

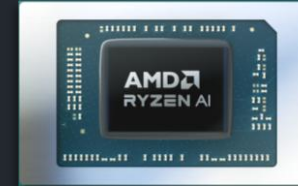


Enterprise

Broad ecosystem of OEM
and solution partners

AMD Instinct™ MI300
Accelerators

5th Gen AMD EPYC™
Processors



PC

The world's most powerful NPU
for next-gen AI PC experiences

AMD Ryzen™ AI 300 Series
Mobile Processors

AMD Ryzen™ AI PRO 300 Series
Mobile Processors



Embedded

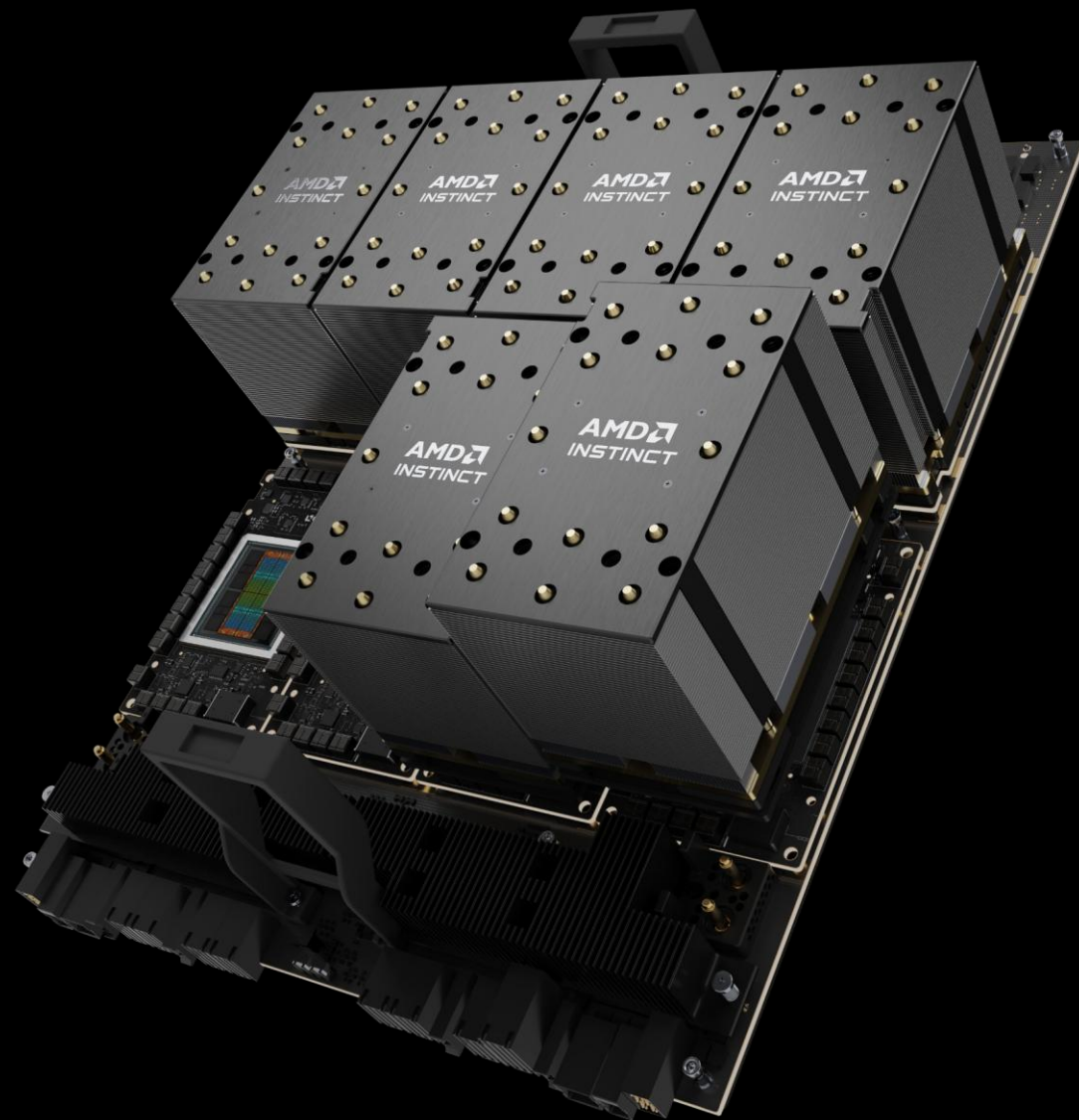
Adaptive SoC + AI
for Embedded

AMD Versal™ AI Edge

AMD Zynq™ MPSoC

Cloud to end point AI software

AMD INSTINCT™ GPU ACCELERATORS



AMD Radeon™ AI PRO R9700

Graphics



128 AI Accelerators

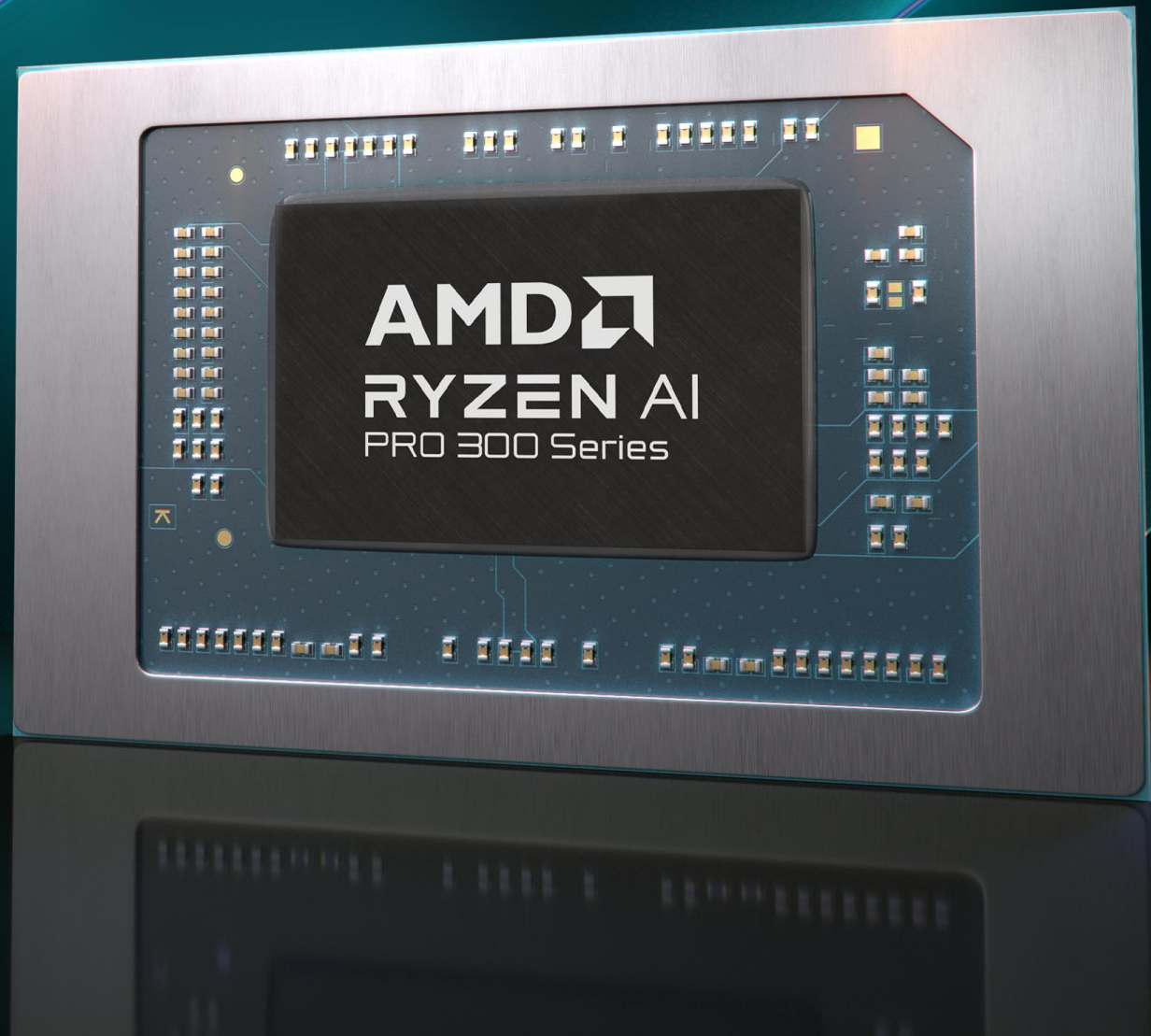
32GB GDDR6

Up to
96 TFLOPS
Peak Half-Precision

Up to
1531 TOPS
INT4 Sparse

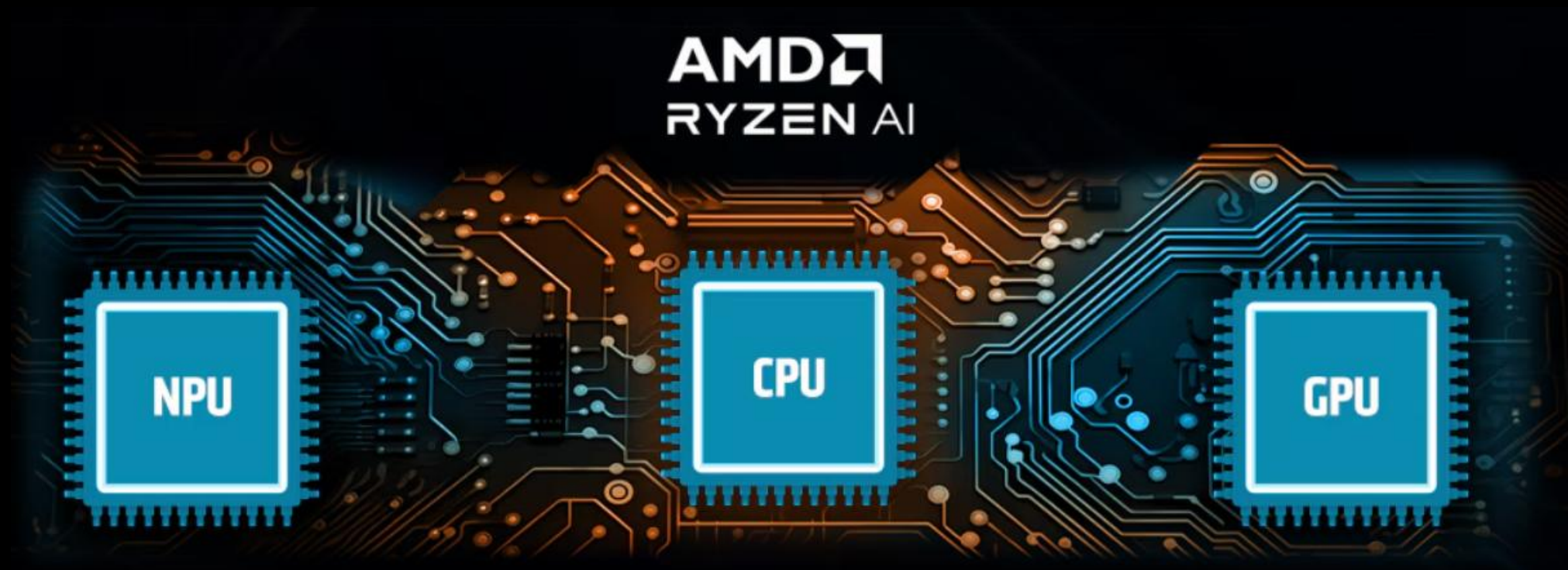
300W TDP

AMD Ryzen™ AI SERIES Processors



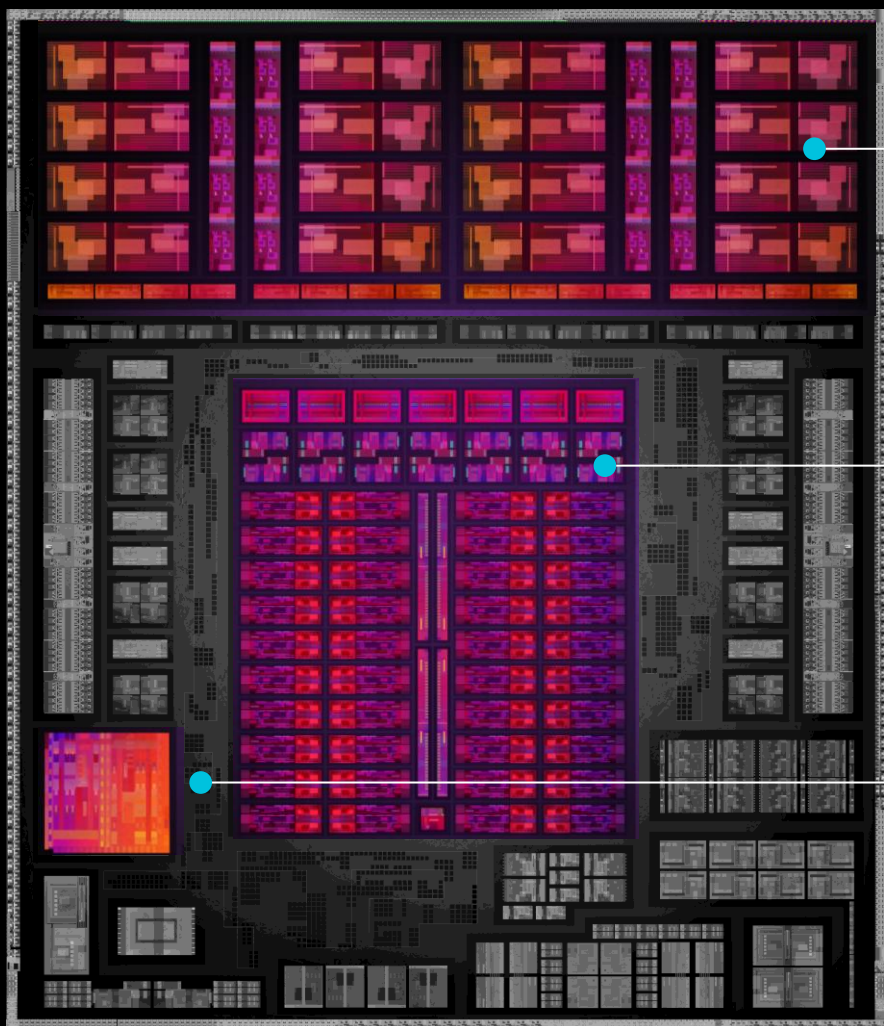
The Definition of AMD Ryzen™ AI PC

An AMD AI PC is a PC that contains an **NPU** (neural processing unit). AMD AI PC will run AI applications on the CPU, GPU, and/or the NPU depending on the characteristics of the task. All three of these components are necessary for a PC to qualify as an AMD AI PC.



AMD Ryzen™ AI Max PRO Series Processors

Redefining Performance For Compact Workstations



Up to **16 Core** CPU
For cutting-edge single- and multi-threaded performance



Up to **40 CU** GPU
For powerful integrated, ISV-certified graphics and GPU compute



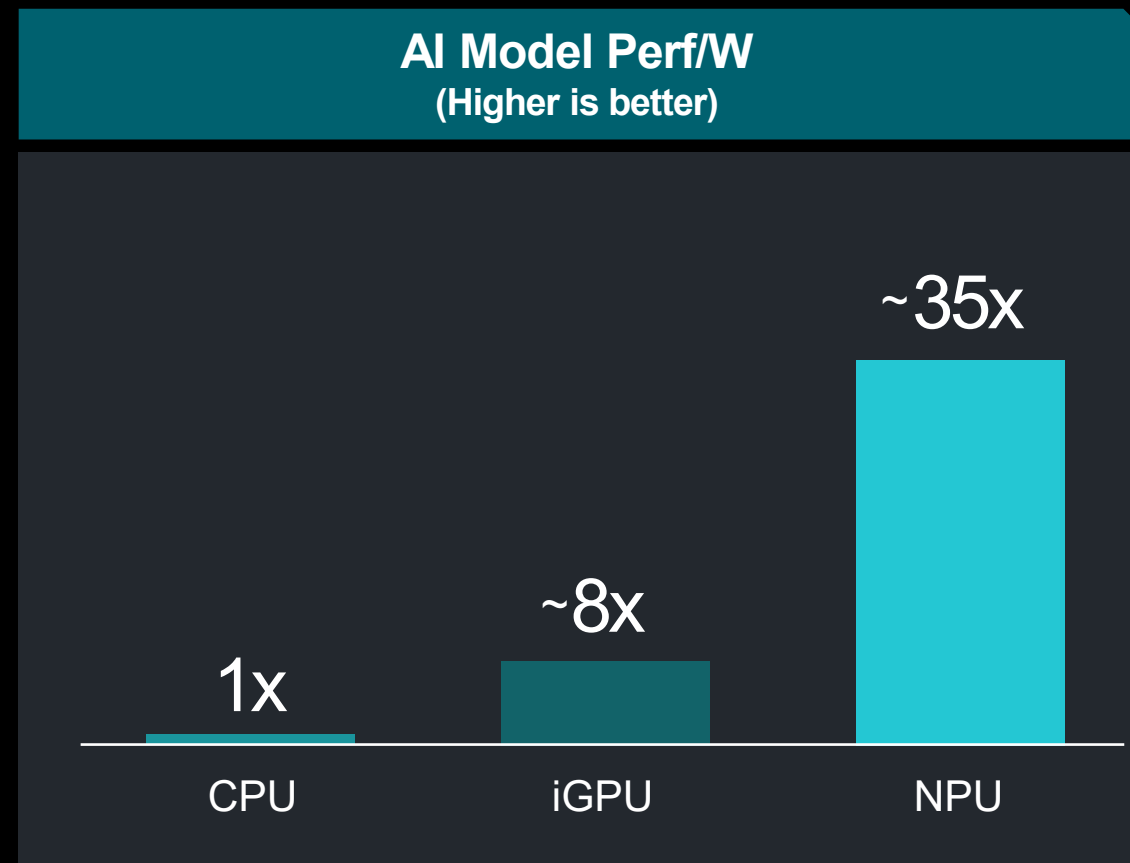
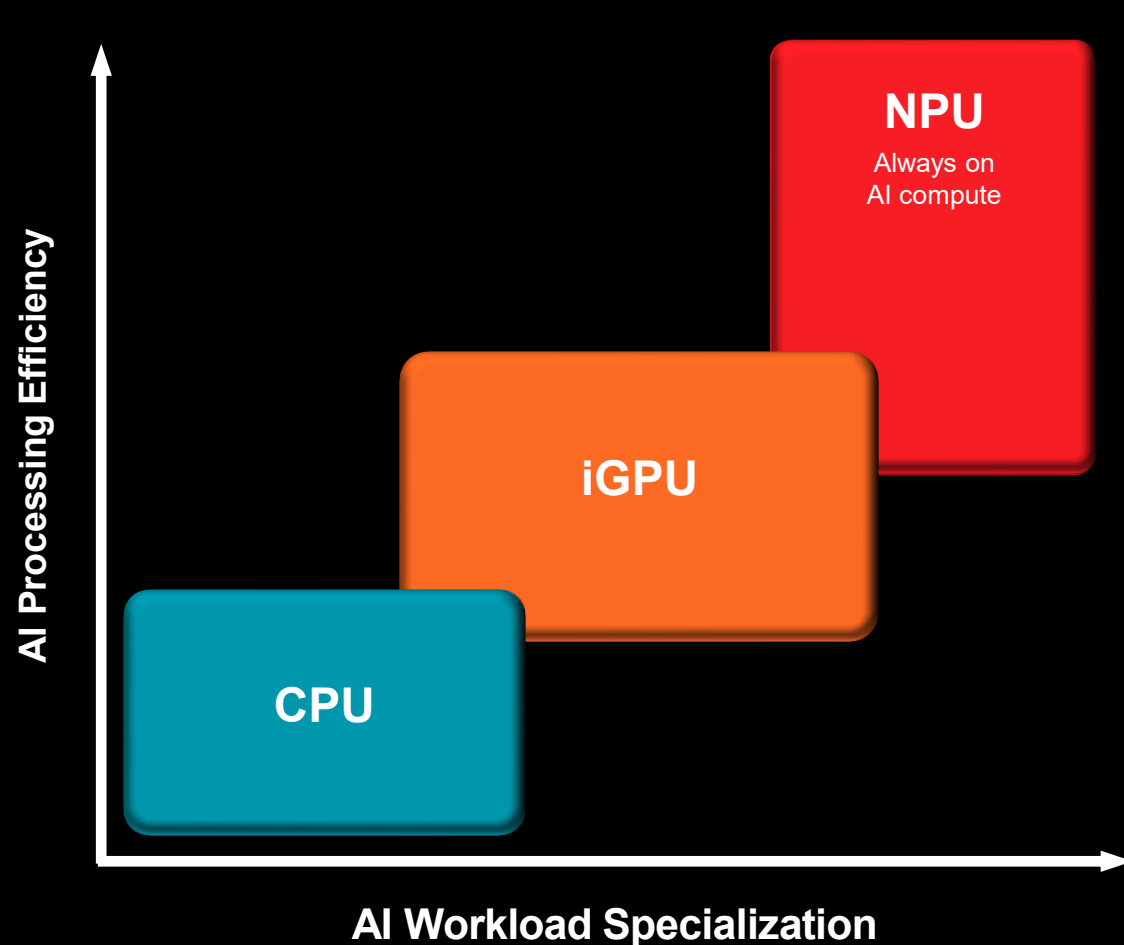
Up to **50+ TOPS** NPU
Energy efficient AI engine for Copilot+PC capabilities*

FEATURING

- ✓ Up to **128GB** unified memory
- ✓ Up to **96GB** GPU memory (VRAM)
- ✓ **256-bit** LPDDR5x memory
- ✓ Up to **32MB** on chip **AMD Infinity Cache™** Technology
- ✓ **AMD PRO** Technologies

WHY IS THE NPU A *MUST-HAVE* FEATURE FOR AI?

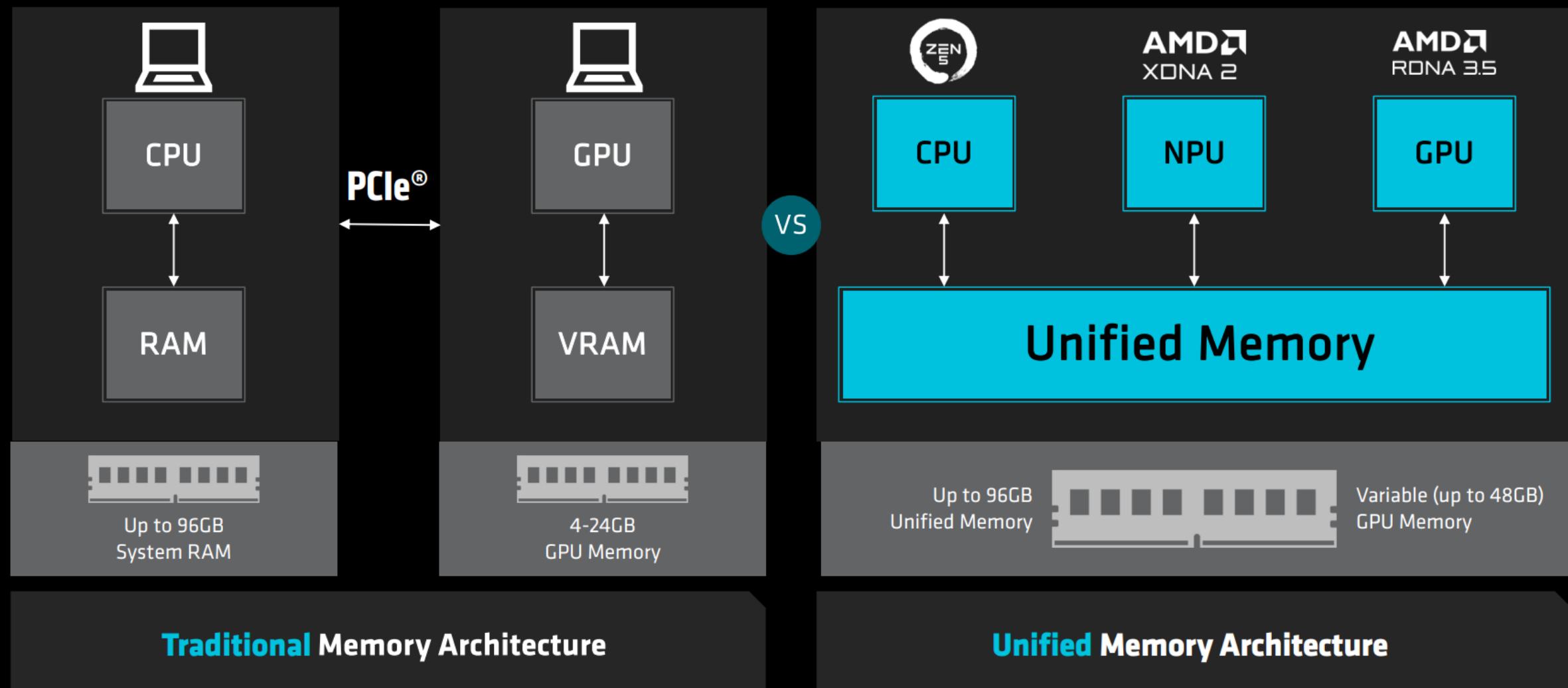
Because AI Engines are the Key to Improving Performance per Watt



NPU is ~4X higher than iGPU

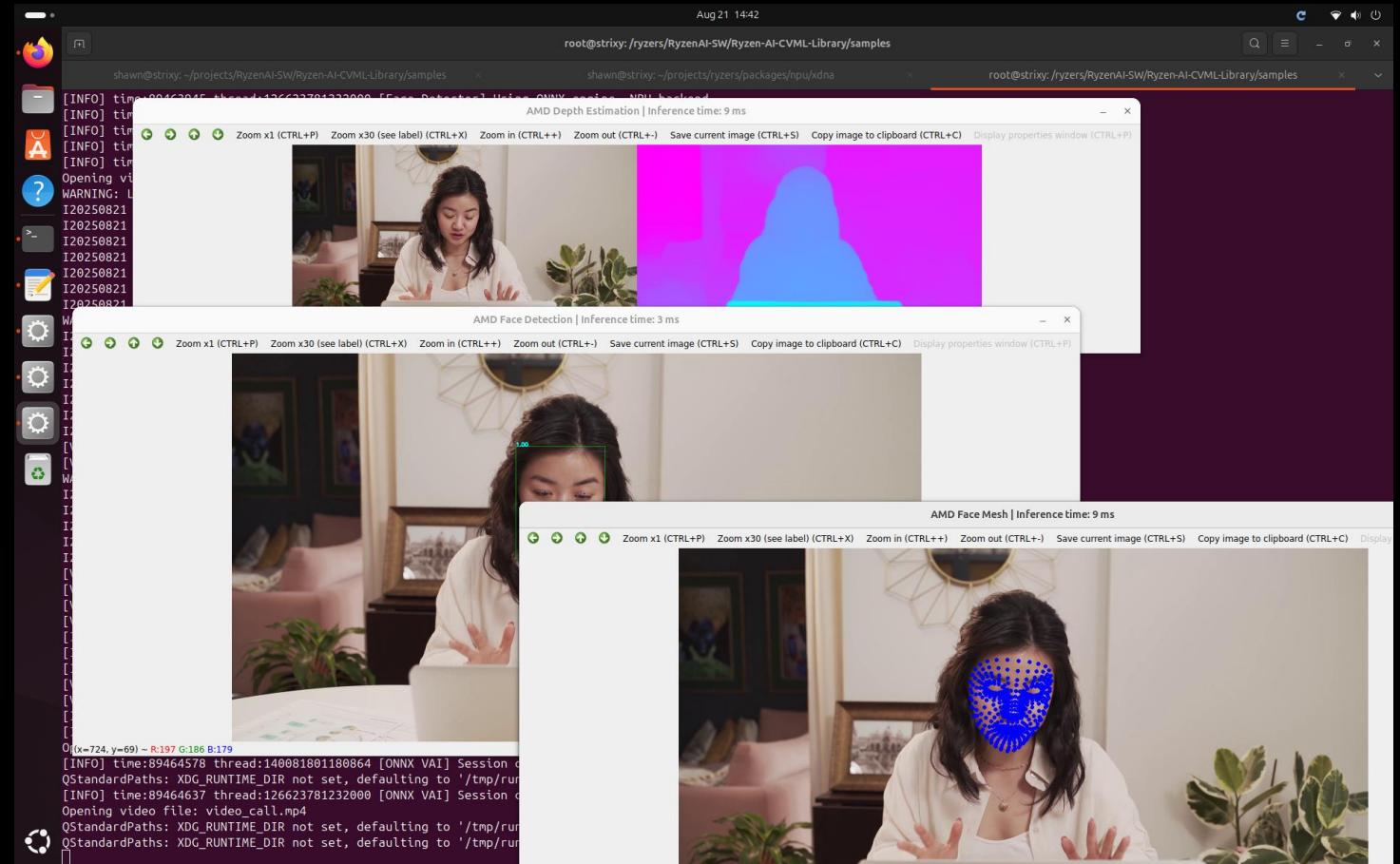
AMD Ryzen™ AI PRO 300 Series Processors for Workstations

Unified Memory Architecture



Ryzen AI CVML library

- C++ library that ships with prebaked binaries for 3 apps:
 - Depth estimation
 - Face detection
 - Face mesh
- Working today within Ryzers



More NPU INFO – Ryzers Repo

AMD Ryzers

Ryzen AI Robotics, Vision and ML Dockerfiles

This repository provides a collection of composable Dockerfiles and build scripts for deploying and running software, full applications, and select demonstrators on AMD Ryzen AI hardware. The project is designed to streamline the setup of AI workloads, robotics, vision, and other applications optimized for Ryzen AI.

Features

- **Verified AMD Support for Popular Frameworks:** A variety of Robotics and ML software frameworks supported across Ryzen AI platforms.
- **Optimized for Ryzen AI:** Includes support for hardware-accelerated AI workloads and accelerators like the iGPU and NPUs.
- **Minimal Host Software Requirements:** Standard Ubuntu support with minimal software requirements
- **Composable Dockerfiles:** Modular design for reusability across different applications.

Supported Packages

Category	Software
LLM	ollama , llamacpp , lmstudio , GEPA , Lemonade
VLM	Gemma3 , Smo1VLM , Phi-4 , LFM2-VL
VLA	OpenVLA , Smo1VLA , GR00T-N1.5 , openpi , CogACT , MolmoAct , MolmoAct2
Graphics	O3DE
Robotics	ROS 2 , Gazebo , LeRobot , ACT , RAI
Simulation	Genesis , PyDrake
Vision	OpenCV , SAM , MobileSAM , ncnn , DINOv3 , SAM3 , Ultralytics
Ryzen AI NPU	XDNA , IRON , NPUEval , Ryzen AI CVML
Adaptive SoCs	PYNQ.remote
Utilities	JupyterLab , amdgpu_top



AMD
ROCm

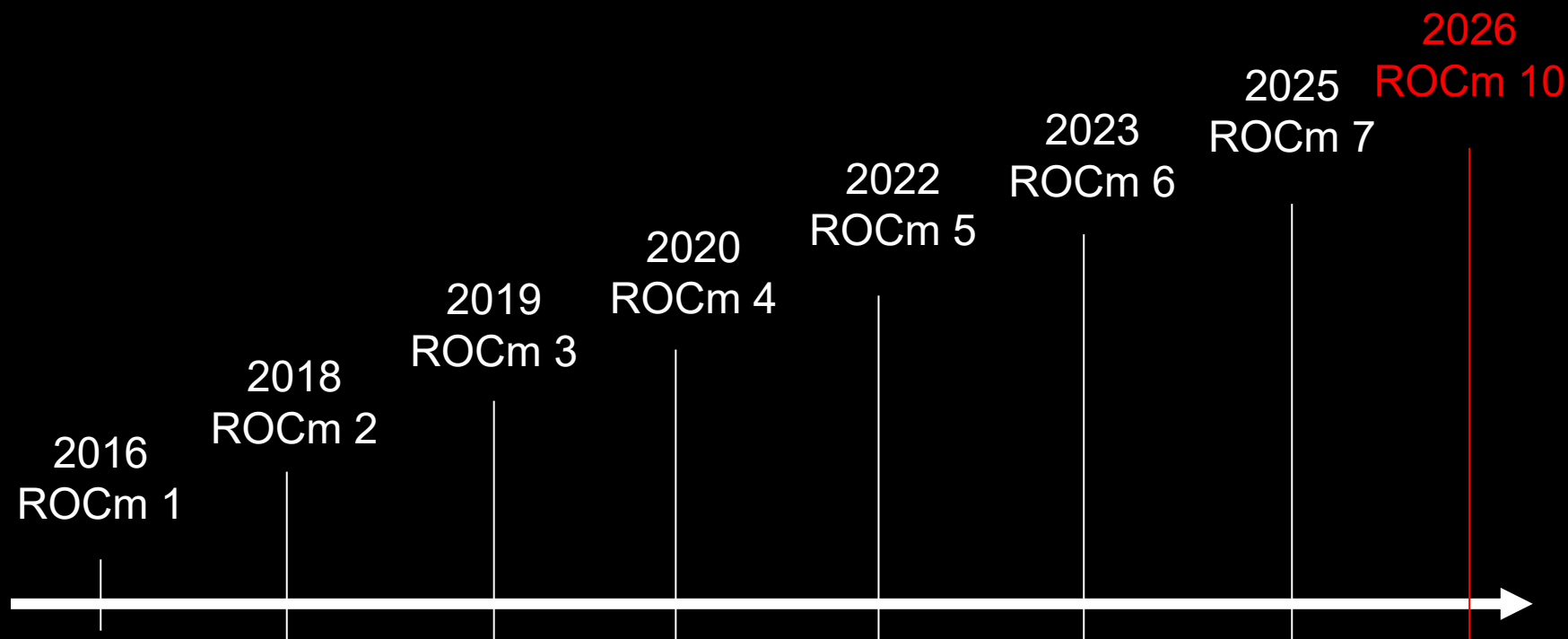
The AMD ROCm logo is displayed in white. It features the word "AMD" in a bold, sans-serif font, followed by a square icon containing a stylized, geometric arrow pointing towards the top-right. Below "AMD" is the text "ROCm" in a similar font style.

Enabling open innovation
at scale

Radeon Open Compute Platform

The first GPGPU software stack from ATI/AMD was Close to Metal, which became Stream.

- ROCm was launched around 2016 with the Boltzmann Initiative¹
- ROCm stack builds upon previous AMD GPU stacks; some tools trace back to GPUOpen and others to the Heterogeneous System Architecture (HSA).



AMD 
ROCm



Comprehensive AI Software Stack

AI Models and Algorithms



Support all major frameworks and models

Libraries

Compilers and Tools

Runtime

Expanded Gen AI optimizations

New algorithms

New libraries

Expanding platform support

AMD GPUs



AMD
INSTINCT



AMD
RADEON

Extended developer support



Deep collaboration with developer community



PyTorch

Day 0 support for latest features



Triton

Vendor agnostic compiler alternative

 vLLM

 SGLang

 JAX

 TensorFlow

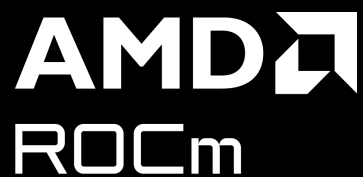
 ONNX Runtime

 OpenXLA

 DeepSpeed

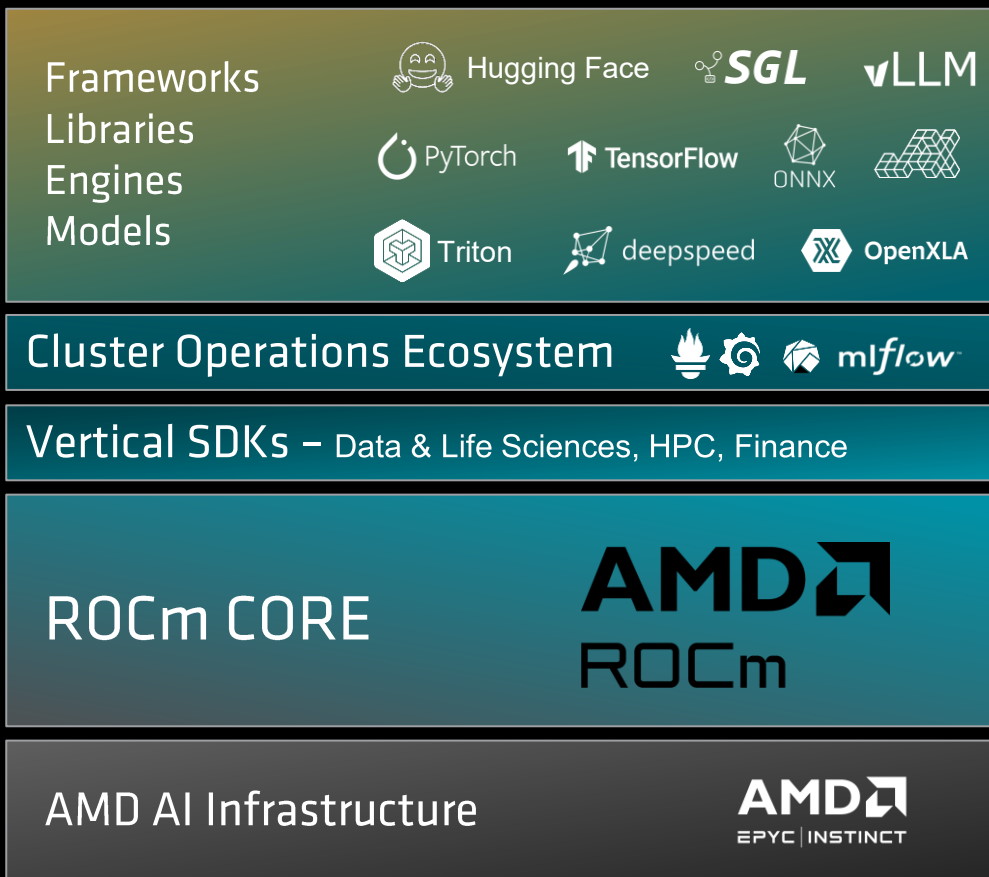
 MLIR | IREE

Increasing open-source contributions and expanding footprint



Comprehensive and Open Infrastructure Stack

A decade of innovation, partnerships and collaboration



Core Math & Compute

rocThrust

hipCUB

hipSPARSE

hipSparseLt

hipFFT

hipBLAS

hipBLASLt

hipRAND

Distributed Compute

RCCL

rocSHMEM

Media I/O

rocDecode

rocJPEG

Programming & Runtime

HIP

OpenMP

LLVM

Profiling & Debugging

rocGDB

rocm-compute

rocm-systems

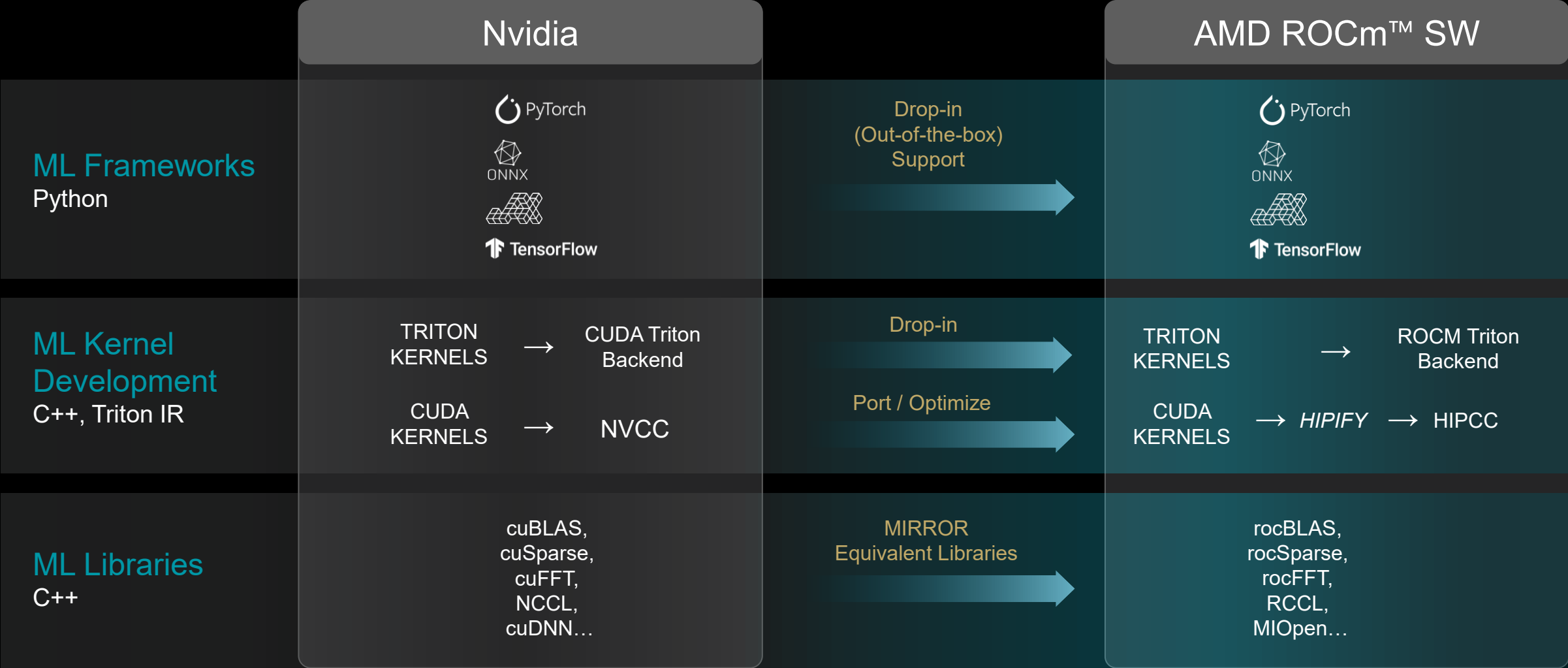
omnistat

Control & Monitoring

rocm-info

amd-smi

Transitioning AI Workloads to AMD GPUs



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ROCm™ Software: Can You Spot a Difference?

NVIDIA CUDA

```
import torch
import torch.nn as nn

# Get cpu or gpu device for training.
device = "cuda:0" if torch.cuda.is_available() else "cpu"
print(f"Using {device} device")

# Define model
class Network(nn.Module):
    def __init__(self):
        super().__init__()
        self.flatten = nn.Flatten()
        self.linear_relu_stack = nn.Sequential(
            nn.Linear(28 * 28, 512),
            nn.ReLU(),
            nn.Linear(512, 512),
            nn.ReLU(),
            nn.Linear(512, 10)
        )

    def forward(self, x):
        x = self.flatten(x)
        logits = self.linear_relu_stack(x)
        return logits

model = Network().to(device)
print(model)
```



AMD ROCm™ Software

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AMD ROCm Documentation

Core SDK

AI Ecosystem

GPU Systems and Infrastructure

Toolkits

ROCm.AI

Blogs

Developer Hub

AMD ROCm 10.0.0

Search Ctrl + K

Release notes

Transition guide

What is ROCm?

Compatibility matrix

Install

Install ROCm

Prerequisites

Installation

Post-installation

Uninstalling

ROCm Runfile Installer

Build ROCm from source

Install AI frameworks

Components

ROCm Core SDK

ROCm HPC SDK

> Install AMD...

Install AMD ROCm 10.0.0

Compare installation methods

Use the following selector to choose your installation method for your supported AMD GPU or APU and operating system. For system requirements and support information, see the [Compatibility matrix](#). To learn more about changes introduced in ROCm 10.0.0, see the [Release notes](#).

Note

If your GPU is not listed, it might be community-enabled through TheRock nightly builds. For more information, see [TheRock supported GPUs](#). For installation guidance, see [TheRock releases](#).

Device family	All	AMD Instinct™	AMD Radeon™	AMD Ryzen™
Use case	Compute	Compute + graphics		
Ryzen APU	AMD Ryzen AI 7 350 (Radeon 860M) (gfx1152)			
Operating system	Ubuntu	Windows		
Ubuntu version	26.04	24.04.4		

Installation environment

Device family

AMD Ryzen

Use case

Compute

Ryzen APU

AMD Ryzen AI 7 350 (Radeon 860M) (gfx1152)

Operating system

Ubuntu

Ubuntu version

24.04.4

Installation method

apt

Contents

Prerequisites

Install the OEM kernel

Install additional packages

Configure permissions for GPU access

Installation

Post installation

Lemonade SDK

The Lemonade SDK makes it easy to run Large Language Models (LLMs) on your PC. Our focus is using the best tools, such as neural processing units (NPUs) and GPU acceleration, to maximize LLM speed and responsiveness.



[Downloading and Getting Started - Lemonade Server Documentation](#)

Do **NOT Change the Password!**

Account: wildbot
Password: wildbot2026

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